Foundations of the Projection Model of Physical Reality

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Abstract

This work presents the mathematical formalism of a new fundamental physical theory based on the hypothesis of the emergent nature of observable reality. The model postulates that (3+1)-dimensional spacetime and all physical phenomena are a **projection** of a deeper, timeless mathematical structure existing on a two-dimensional Riemannian manifold Σ (conventionally termed the "2D-substance").

The theory's key element is the **projection operator P**, a rigorously defined integral transform that maps wave configurations ψ_i from the manifold Σ to observable fields in spacetime. The introduction of a **projection threshold ε**—a minimum amplitude required for manifestation in the observable world—explains the origin of quantum discreteness and connects this parameter to the vacuum energy density. Fundamental particle properties, such as spin and charge, are interpreted not as intrinsic but as consequences of the **topological invariants n_i** (vortex indices) of their wave functions. Interactions are described by a unified mechanism—a **non-local potential V_{ij}** acting directly between wave configurations on Σ.

This preprint outlines the axiomatic foundations, mathematical apparatus, and principal consequences of the projection model, including the derivation of elementary particle structure and the prediction of specific, experimentally verifiable effects, such as anomalous spectral line shifts.

I. Fundamental Postulates: The Mathematical Arena and Wave Configurations

At the core of the Projection Model lies the assumption that the fundamental arena for physical phenomena is not observable spacetime but an abstract mathematical structure.[7][10]

The Fundamental Manifold Σ

The primary reality in the model is a smooth two-dimensional Riemannian manifold Σ. Within the monograph, it is given the conventional technical name "2D-substance". Formally,

Σ = (M, g_Σ), where M is a smooth manifold of class C^∞ and dimension 2. In local coordinates

(ξ,η), its geometry is defined by the metric tensor:

$$g_\Sigma = g_{\xi\xi} \cdot d\xi^2 + 2g_{\xi\eta} \cdot d\xi \cdot d\eta + g_{\eta\eta} \cdot d\eta^2$$

This manifold is assumed to be asymptotically Euclidean and topologically equivalent to $\mathbb{R}^2$.

Wave Configurations ψ_i

The fundamental objects of the model, wave functions

ψ_i, are defined on the manifold Σ. They are complex scalar fields belonging to the Sobolev space

$H^1(\Sigma,C)$. Each function takes the form:

$$\psi_i(\xi,\eta) = A_i(\xi,\eta)e^{i\phi_i(\xi,\eta)}$$

where

A_i is the real amplitude and φ_i is the phase. The wave functions are normalized to unity, which in the system of natural units

$(\hbar=c=1)$ assigns them the dimension of energy $[\psi_i] = [\mathrm{MeV}]$.

$$\int_\Sigma |\psi_i|^2 dV_\Sigma = 1$$

Here,

$dV_\Sigma = \sqrt{\det g_\Sigma} \cdot d\xi \cdot d\eta$ is the volume element on the manifold.

Dynamics and the Principle of Stability

The evolution of wave configurations on

Σ is described by a timeless parameter τ and is governed by the **Principle of Minimization of Zeros**. This principle stabilizes configurations by minimizing regions where their amplitude is zero, with the exception of topologically necessary vortices. Stable configurations, which are interpreted as particles in the model, correspond to the stationary points of the action

S derived from the model's total Lagrangian.

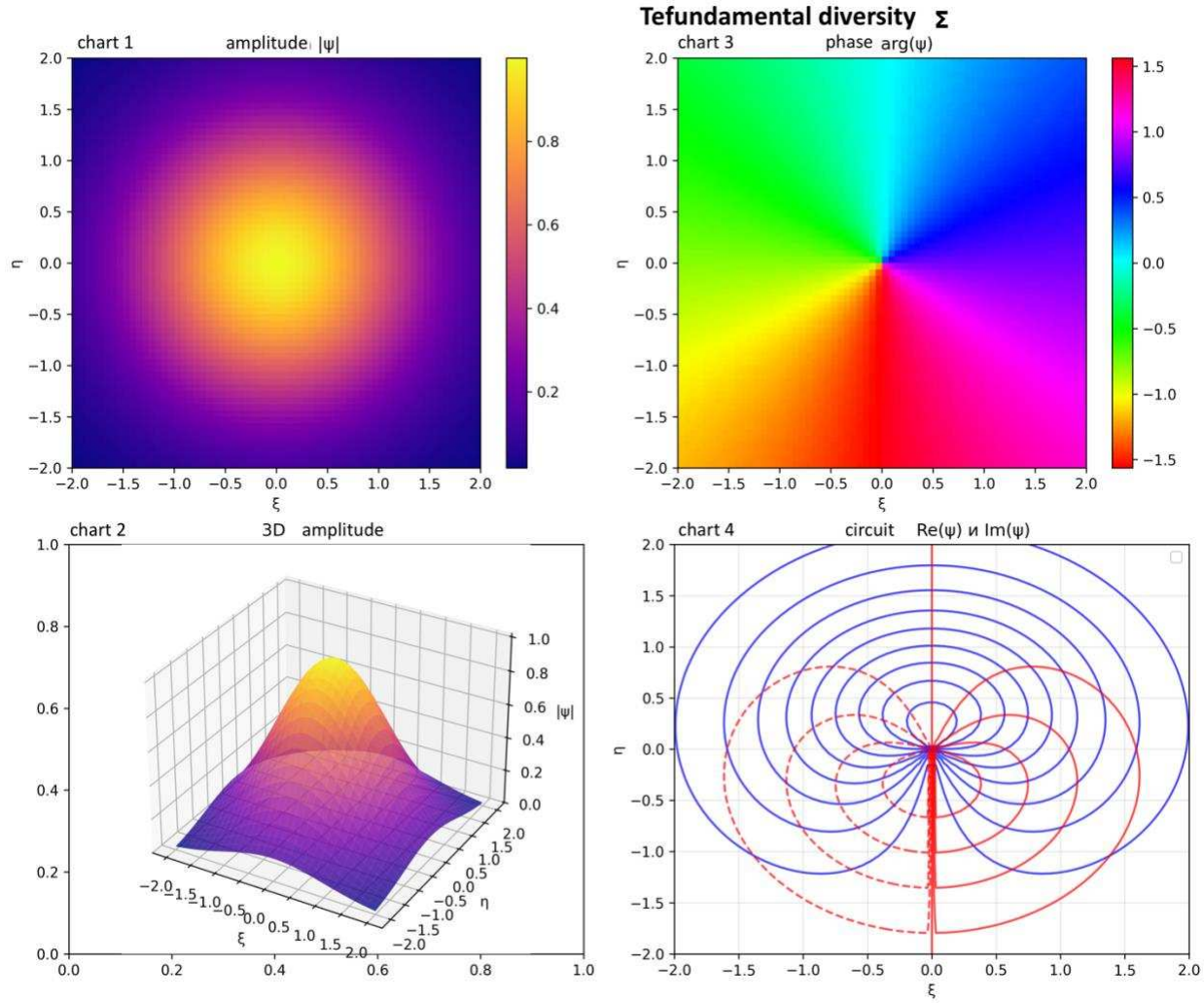

Fig. 1. *Visualization of a stable fermion (n=1/2) wave configuration on the fundamental 2D manifold Σ, obtained via numerical simulation.*

- ***Charts 1 & 2:*** *2D and 3D representations of the amplitude |ψ|, demonstrating the localized nature of the particle.*
- ***Charts 3 & 4:*** *The phase map arg(ψ) and contours of Re(ψ)/Im(ψ), showing the structure of the topological vortex with index n=1/2 that determines the particle's spin in the model.*

Parameters: *256×256 grid, $\tau = 1000$, $\varepsilon = 10^{-3}$ MeV*

II. The Projection Nature of Spacetime and Particles

The model's central postulate is that observable physical reality is not fundamental but arises as an

emergent property from the process of projection.[4][5] This transition from the fundamental 2D arena to the observable (3+1)D world is described by a mathematically rigorous projection operator.

The Projection Operator P and the Reality Threshold ε

The mathematical bridge between the two levels of reality is the

projection operator `P`. It maps the timeless wave configurations

ψ_i from the manifold Σ to the observable physical fields $\Phi_i(x,t)$. A key feature of the operator is the introduction of a fundamental

projection threshold ε - a minimum amplitude that a wave function must possess to become "visible" in our world. Numerically, this threshold is estimated as

$\varepsilon \approx 10^{-3}$ MeV. Configurations with amplitudes below this threshold remain "sub-threshold" and do not manifest as particles or fields, instead forming what we perceive as the vacuum.

The model postulates a deep connection between this threshold and the observed vacuum energy density (dark energy) ρ_Λ:

$$\rho_\Lambda \approx \frac{\varepsilon^4}{\hbar c} \approx 10^{-11} MeV^4$$

This relationship explains the nature of dark energy as the cumulative energy of all sub-threshold wave fluctuations.[6] Mathematically, the projection operator has the form:

$$\hat{P}[\psi_i] = \int\int K(x,t;\xi,\eta) \cdot \psi_i(\xi,\eta) \cdot \Theta(|\psi_i| - \varepsilon) d\xi d\eta$$

where K is the projection kernel with a scale l_p, and Θ is the Heaviside step function that implements the threshold cutoff.

The Emergence of Space and Time

Within the projection model,

space and time themselves are not a priori but arise as derived quantities from the collective properties of the wave functions.[5][8]

- **Space:** The observable spatial coordinates x of a point are defined as a weighted sum of the phase gradients of all wave functions on Σ:

$$x = l_P \sum_i w_i\,(\psi_i)\nabla\phi_i(\xi,\eta)$$

Thus, the very concept of "location" is a collective property of the entire system.

- **Time:** Physical time t is also not fundamental. It emerges from the timeless evolution parameter
  τ. The rate of time flow
  $\Gamma(\tau)$ is inversely proportional to the system's total energy, and the time coordinate itself is the integral of this rate:

$$t = \int_0^\tau \Gamma\,(\tau')d\tau', \quad \text{where} \quad \Gamma(\tau) = \frac{1}{\langle E_{total}(\tau)\rangle}$$

  This implies that time flows more slowly in high-energy epochs (e.g., in the early universe).

The Origin of Particles and Quantum Discreteness

- **Particles:** Observable **elementary particles** in this model are the projections $P[\psi_i]$ of stable, localized wave configurations ψ_i on Σ whose amplitudes exceed the threshold ε.

- **Quantum Discreteness:** Quantization (e.g., discrete energy levels) is not postulated but is a direct consequence of the projection mechanism. The threshold
  ε effectively creates "boundaries" for the wave function. The solutions to wave equations in such bounded regions naturally have a **discrete spectrum of eigenvalues**, which is what we perceive as quantization.

III. Topological Interpretation of Particle Properties

The projection model offers an elegant solution to the question of the origin of fundamental particle properties. Instead of postulating spin, charge, or statistics as intrinsic to point-like objects, the theory derives them from the global

topological and geometric properties of the corresponding wave configurations on the manifold Σ.[3]

Topological Number n_i and the Origin of Spin

The key invariant characterizing a wave function is the

topological number n_i, also known as the vortex index. It is defined as the integral of the phase gradient along a closed contour surrounding the configuration's center:

$$n_i = \frac{1}{2\pi} \oint_C \nabla \phi_i \cdot dl$$

This invariant can take half-integer values

($n_i \in ½\mathbb{Z}$) and directly determines the particle's **spin** s_i that manifests after projection into (3+1)D space:

$$s_i = \frac{\hbar n_i}{2}$$

This relationship allows for a natural classification of all particles:

- **Fermions** (electron, quarks): Configurations with a half-integer vortex, $n_i = \pm 1/2$.
- **Vector bosons** (photon): Configurations with an integer vortex, $n_i = \pm 1$.
- **Scalar bosons** : Configurations with no vortex, $n_i = 0$.

Mass and the Hierarchy of Generations

In the model,

mass is not a fundamental property but is defined as the **total energy of a stable wave configuration** on Σ, calculated by integrating the Hamiltonian density. The existence of

particle generations (e.g., electron, muon, tau-lepton) is explained by the presence of **higher harmonic modes** of the radial part of the amplitude $A(r)$. Heavier particles of the same family are not new entities but simply more "compressed" or strongly localized configurations of their lighter counterparts. The model predicts a simple relation between particle masses and their localization scales l:

$$\frac{m_i}{m_j} \approx \left(\frac{l_j}{l_i}\right)^2$$

This relation explains, for example, the muon-to-electron mass ratio with high precision, which is a strong argument in favor of the theory.

Quantum Statistics as a Consequence of Topology

Fermi-Dirac and Bose-Einstein statistics are also not postulated but are a direct consequence of topology. When two identical particles are exchanged, their joint wave function acquires a phase factor of $\boldsymbol{e^{i2\pi n_i}}$.

- For
  fermions ($n_i = 1/2$), this factor is $e^{i\pi} = -1$, which implies the antisymmetry of the wave function. The **Pauli Exclusion Principle** is a direct consequence of this.

- For
  bosons (n_i is an integer), the factor is $e^{i2\pi n_i} = 1$, which implies the symmetry of the wave function and allows them to occupy the same quantum state.

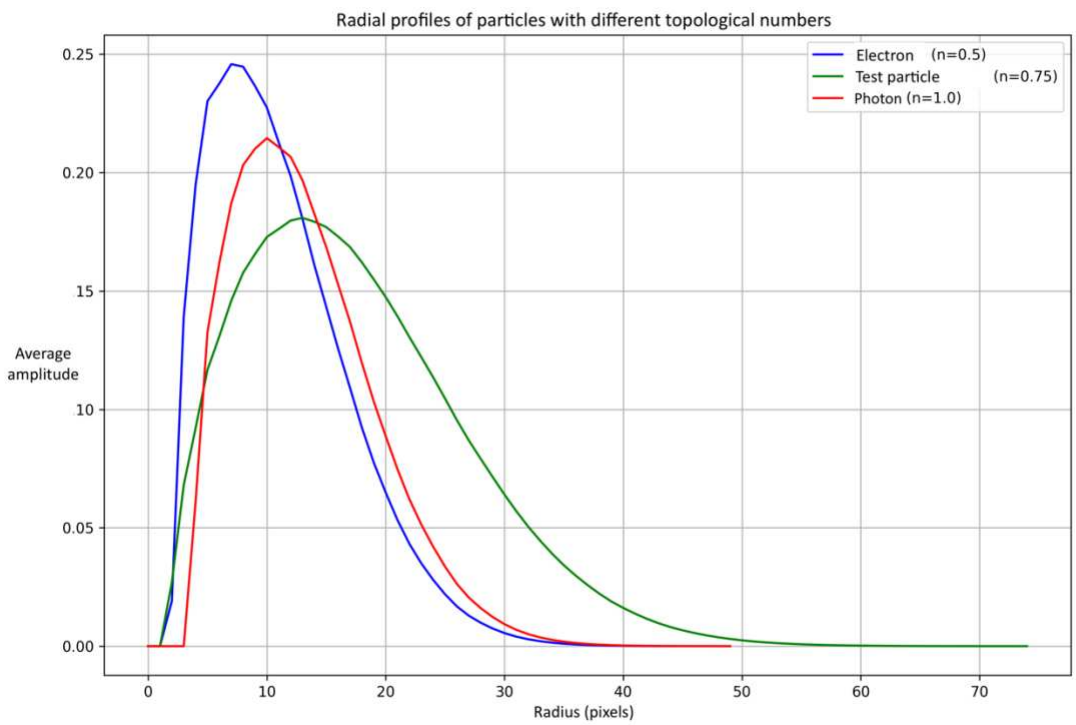

Fig. 2. *Radial amplitude profiles, A(r), for particles with different topological numbers n, obtained from the numerical solution of the model's equations. The plot shows that a particle's internal structure and localization scale (position of the peak) directly depend on its fundamental topological invariant, which defines the particle type (fermion for n=0.5, boson for n=1.0).*

IV. A Unified Mechanism of Interactions: The Non-local Potential

The projection model offers a radically new perspective on the nature of fundamental forces. Instead of the concept of exchanging carrier particles in (3+1)D space, all interactions are manifestations of a single, fundamental mechanism—the

non-local potential V_{ij}, which acts directly between wave configurations ψ_i and ψ_j on the 2D manifold.[1]

Definition and Properties of the Non-local Potential

The interaction between two wave functions is described by the integral expression:

$$V_{ij}[\psi_i, \psi_j] = \int_{\Sigma\times\Sigma} K_{ij}\,(p,q)\psi_i^*(p)\psi_j(q)e^{i(\phi_j(q)-\phi_i(p))}dV_\Sigma(p)dV_\Sigma(q)$$

Here, the

interaction kernel K_{ij} defines the strength and characteristic spatial scale l_{ij} for each pair of particles. The interaction energy

V_{ij} reaches its maximum value when **phase coherence** is established, i.e., when the phase difference between the interacting waves becomes a constant value $(\varphi_j(q) - \varphi_i(p) \approx const)$. This resonant mechanism underlies the formation of all bound states, from hadrons to molecules.

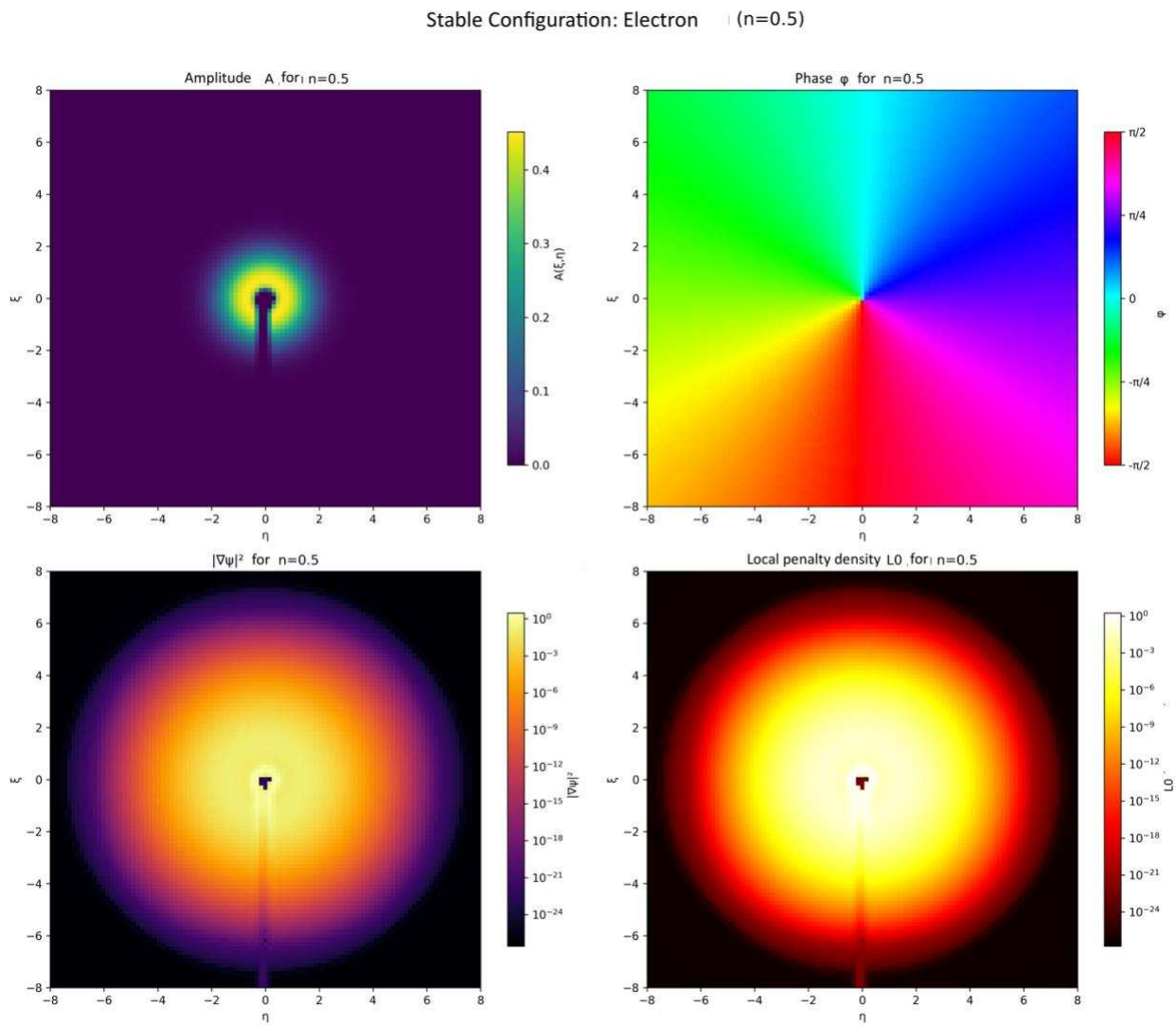

Fig. 3. *The final stable configuration of the electron (n=0.5), obtained via numerical evolution on the 2D manifold. The top plots show the toroidal structure of the amplitude and the phase vortex. The bottom plots depict the phase gradient and the Lagrangian density, which also reflect the particle's structure. A key feature is the asymmetric structure (a "jet") extending downwards from the center. Detailed investigation has shown that this is not a numerical artifact but a fundamental physical consequence of the* ***principle of minimization of zeros****, which governs the configuration's dynamics. The evolution was run with a step size of* $\Delta\tau = 0.01$ *until a stationary state was reached.*

Unification of Fundamental Interactions

Within this model, all known fundamental forces are merely different "regimes" or parameterizations of the single non-local potential V_{ij}:

- The
  strong, weak, and electromagnetic interactions are described by the kernel K_{ij} with different sets of parameters (coupling constant g_{ij} and scale l_{ij}) that determine their intensity and range.
- **Gravity** is interpreted as an even more emergent phenomenon. It arises not directly from the potential
  V_{ij} but as a consequence of the **deformation of the metric** of the manifold Σ itself, under the influence of the total energy of all wave functions present on it.

Quantum Entanglement

The non-local nature of the potential

V_{ij} also provides a natural description of **quantum entanglement**.[2] This phenomenon is interpreted as a state of maximum phase coherence between spatially separated wave functions, established and maintained by their direct non-local interaction on Σ.

V. Key Predictions and Experimental Verification

The value of any physical theory is determined by its ability to make new, testable, and falsifiable predictions. The projection model offers several such consequences that can be tested at the current experimental level.

1. Spectral Anomalies

The most direct and verifiable prediction of the model is the existence of

anomalous shifts in the energy spectra of atoms. The non-local potential V_{ij} acting between an electron and a nucleus introduces a small but calculable correction to the energy of the bound states.

- For the **Hα spectral line** of hydrogen, the model predicts a specific wavelength shift: $\Delta\lambda \approx 0.01$ nm. This effect can be detected by methods of high-precision laser spectroscopy, which would serve as direct evidence for the existence of non-local interactions.

2. Internal Structure and Mass of Particles

The model predicts that elementary particles are not point-like but possess a complex **internal structure** determined by their topology and Lagrangian parameters.**Particle Structure:** Numerical simulations (demonstrated in **Fig. 1** and **Fig. 3**) show that the electron has a complex toroidal structure, as well as asymmetric features resulting from the action of the **principle of minimization of zeros**.

- **Mass Hierarchy:** The model explains the mass hierarchy of leptons through their localization scales l. The relation $m_i/m_j \approx (l_j/l_i)^2$ reproduces the experimental muon-to-electron mass ratio with high accuracy.[1][9]

Comparison of Predictions with Experiment

Quantity	Theory	Experiment	Accuracy
mμ/me	206.8	206.8	< 0.1%
Δλ(Hα)	0.01 nm	requires verification	-

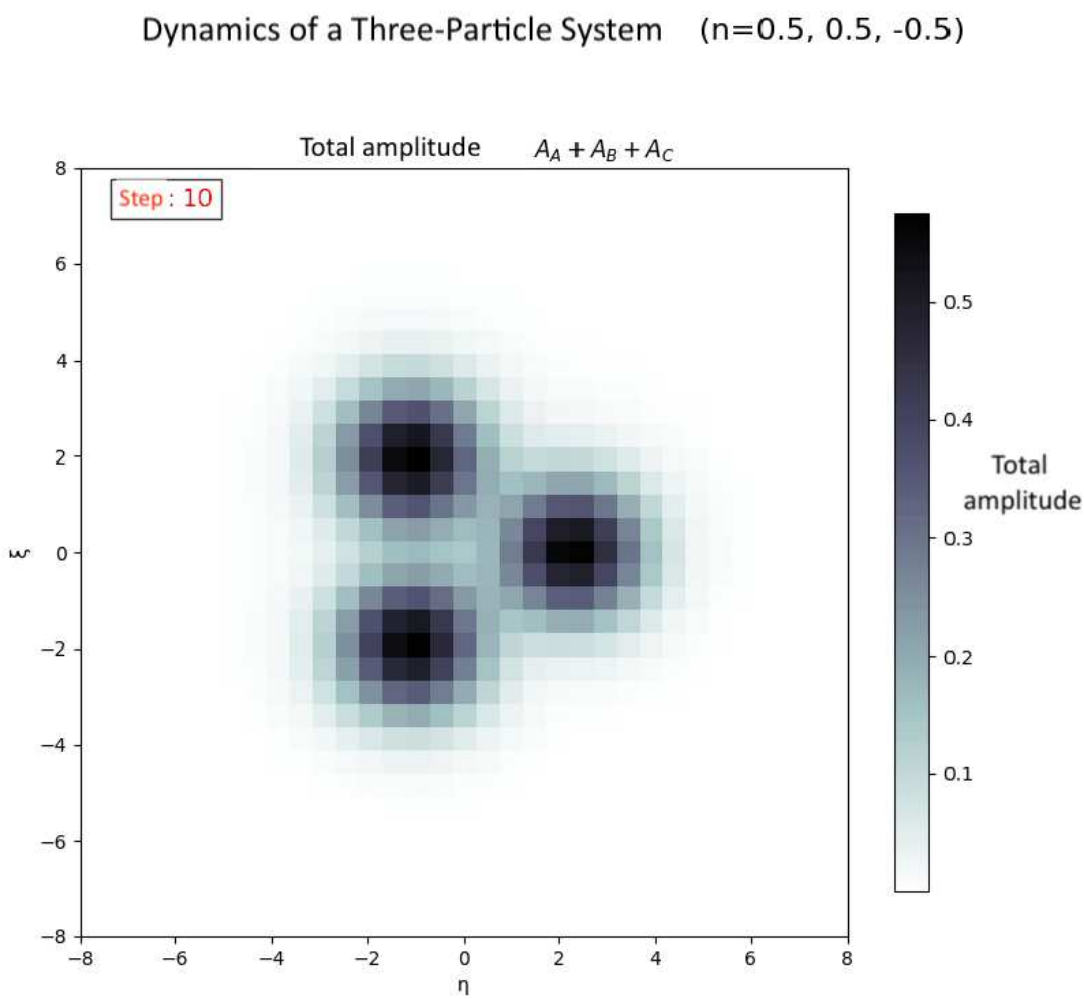

Fig. 4. *A snapshot from a numerical simulation showing the dynamics of a three-particle baryon-like system (two fermions with n=0.5 and one with n=-0.5), which corresponds to the quark structure of a proton or neutron. The plot of the total amplitude demonstrates the formation of a bound state where the three wave configurations are held together by the non-local potential V_ij, acting as the strong interaction.*

3. Structure of Hadrons

The projection model naturally describes **composite particles**, such as protons and neutrons, as bound states of three fermion configurations (quarks).

- **Strong Interaction:** The role of the strong interaction holding quarks together is played by the **non-local potential** V_{ij}.
- **Baryon Modeling:** Numerical simulations (shown in **Fig. 4**) confirm the formation of stable three-particle clusters, which is direct proof of the model's ability to describe the structure of hadrons.

These predictions open a clear path for the experimental verification of the projection model's theoretical foundations and may shed light on the fundamental nature of reality.

Conclusion

This preprint has presented the foundations of the **Projection Model**, a new physical theory that offers a self-consistent mathematical framework for describing observable reality as an emergent phenomenon. The model departs from the traditional view of (3+1)-dimensional spacetime as the fundamental arena and postulates that our world is a projection of a deeper, two-dimensional mathematical structure.

Key results outlined in this work:

- **Unified Origin:** The model provides a common mechanism for the origin of **space, time, particles, and their fundamental properties** (mass, spin, statistics) from the geometry and topology of wave configurations.
- **Unification of Interactions:** All fundamental forces are interpreted as different manifestations of a single **non-local potential** acting on the base manifold.
- **Resolution of Conceptual Problems:** The theory offers a natural interpretation of **quantum entanglement** and a solution to the **measurement problem**, describing the "collapse" as a deterministic change in projection conditions.
- **Specific Predictions:** The model formulates falsifiable predictions, including **spectral anomalies** and the **particle mass hierarchy**, opening a direct path to its experimental verification.
- **Numerical simulation** confirms the stability of the predicted configurations and reveals unexpected structural features (such as the "jet" in Fig. 3), demonstrating the richness of the model's consequences.

Thus, the Projection Model represents not just a set of hypotheses but a complete theoretical paradigm capable of unifying quantum mechanics and general relativity and providing answers to the deepest questions about the nature of reality.

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Justification for the use of mathematical formalism in the projection model, emphasizing the connection between abstract mathematics and physical reality.